

CS 20 Probset 2 Code Snippets

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1. Question 3

1.1. Code

```

import sympy as sp
from sympy import Matrix

raw = \
    11 11 11
    1 0 0 0 0 -1 0 -1 0 0 0 0 0
    0 1 0 1 1 0 0 0 0 0 0 0 0
    0 0 0 0 0 0 0 1 -1 0 0 1 0
    0 0 0 0 0 0 0 0 0 -1 1 -1 0
    0 0 0 0 -1 0 1 0 0 0 -1 0 0
    -1 -1 1 0 0 0 0 0 0 0 0 0 0
    0 0 -1 -1 0 1 -1 0 1 1 0 0 0
    -20 0 -50 0 0 -50 0 0 0 0 0 0 0
    0 10 50 0 0 0 0 0 0 0 0 0 -1
    0 0 0 0 -10 0 -30 0 0 0 0 0 1
    0 0 0 0 0 50 0 -10 0 0 0 0 -5
    0 0 0 0 0 0 30 0 0 30 20 0 0
    0 0 0 0 0 0 0 0 30 0 -10 -5
    11 11 11

def convert_string_to_matrix(raw):
    return [
        list(map(int, line.split()))
        for line in raw.strip().splitlines()
    ]

M = Matrix(convert_string_to_matrix(raw))

rref_matrix, pivot_cols = M.rref()
sp.pprint(rref_matrix)

A = M[:, :-1]
b = M[:, -1]

print("A shape:", A.shape)
print("rank(A):", A.rank())
print("nullity:", A.cols - A.rank())

print("rank([A|b]):", M.rank())

currents = []
for idx, val in enumerate(rref_matrix[:, -1], 1):
    currents.append(val)
    print(f"I_{str(idx).zfill(2)} = {'%.4f'%float(val*1000)} mA")

print(currents)

resistor_values = {
    'A': 10,
    'B': 10,
    'C': 10,
    'D': 50,
    'E': 10,
    'F': 50,
    'G': 30,
    'H': 10,
    'I': 30,
    'J': 10,
    'K': 10,
    'L': 10,
}

resistor_map = {
    'A': 1,
    'B': 2,
    'C': 1,
    'D': 3,
    'E': 5,
    'F': 6,
    'G': 7,

```

```

'H': 8,
'I': 10,
'J': 11,
'K': 12,
'L': 11,
}

for r, curr_idx in resistor_map.items():
    print(r, '%.4f'%float(abs(currents[curr_idx-1]*resistor_values[r])))

```

1.2. Output

```

[1 0 0 0 0 0 0 0 0 0 0 0 3/22 ]
|
|
| -43 |
| 0 1 0 0 0 0 0 0 0 0 0 0 0 — |
| 330 |
|
|
| 0 0 1 0 0 0 0 0 0 0 0 0 1/165 |
|
|
| 1163 |
| 0 0 0 1 0 0 0 0 0 0 0 0 0 — |
| 4620 |
|
|
| -17 |
| 0 0 0 0 1 0 0 0 0 0 0 0 0 — |
| 140 |
|
|
| 0 0 0 0 0 1 0 0 0 0 0 0 0 -2/33 |
|
|
| 1/140 |
|
|
| 13 |
| 0 0 0 0 0 0 0 1 0 0 0 0 0 — |
| 66 |
|
|
| 1933 |
| 0 0 0 0 0 0 0 0 1 0 0 0 0 — |
| 4620 |
|
|
| -13 |
| 0 0 0 0 0 0 0 0 0 1 0 0 0 — |
| 140 |
|
|
| 9/70 |
|
|
| 31 |
| 0 0 0 0 0 0 0 0 0 0 1 0 0 — |
| 140 |
|
|
| 0 0 0 0 0 0 0 0 0 0 0 0 0 ]
A shape: (13, 12)
rank(A): 12
nullity: 0
rank([A|b]): 12
I_01 = 136.3636 mA
I_02 = -130.3030 mA
I_03 = 6.0606 mA
I_04 = 251.7316 mA
I_05 = -121.4286 mA
I_06 = -60.6061 mA
I_07 = 7.1429 mA
I_08 = 196.9697 mA
I_09 = 418.3983 mA
I_10 = -92.8571 mA
I_11 = 128.5714 mA
I_12 = 221.4286 mA
I_13 = 0.0000 mA
[3/22, -43/330, 1/165, 1163/4620, -17/140, -2/33, 1/140, 13/66, 1933/4620, -13/140, 9/70, 31/140, 0]
A 1.3636
B 1.3030
C 1.3636
D 0.3030

```

E	1.2143
F	3.0303
G	0.2143
H	1.9697
I	2.7857
J	1.2857
K	2.2143
L	1.2857